Chapter 9

Chemical Quantities

Information Given by Chemical Equations

- These chemical equations tell us the identities (formulas) of the reactants and products and also show how much of each reactant and product participates in the reaction.
- The numbers (coefficients) in the balanced chemical equation enable us to determine just how much product we can get from a given quantity of reactants.
- The coefficients in a balanced equation give us the relative numbers of molecules.
  - That is, we are interested in the ratio of the coefficients, not individual coefficients.

Methanol Production

Balanced: $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$

Table 9.1 Information Conveyed by the Balanced Equation for the Production of Methanol

$CO(g)$	+	$2H_2(g)$	→	$CH_3OH(l)$
1 molecule CO	+	2 molecules H_2	→	1 molecule CH_3OH
1 dozen CO molecules	+	2 dozen H_2 molecules	→	1 dozen CH_3OH molecules
6.022×10^{23} CO molecules	+	$2(6.022 \times 10^{23})$ H_2 molecules	→	6.022×10^{23} CH_3OH molecules
1 mol CO molecules	+	2 mol H_2 molecules	→	1 mol CH_3OH molecules

Chemical Equation Practice

- Given the following equation: $2Al_2O_3 \rightarrow 4Al + 3O_2$

How many molecules of oxygen can be formed from 4 molecules of Al_2O_3?

If 8 atoms of Al are produced, how many molecules of oxygen will also be produced?

Using Balanced Equations

- We can use an equation to predict the moles of products that a given number of moles of reactants will yield.
- For example, consider the decomposition of water to give hydrogen and oxygen, which is represented by the following balanced equation:

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

- This equation tells us that 2 moles of H_2O yields 2 moles of H_2 and 1 mole of O_2.

Mole Ratios

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

- We can obtain the following mole ratios from this balanced equation:

$$2 \text{ mol } H_2O = 2 \text{ mol } H_2$$

$$2 \text{ mol } H_2O = 1 \text{ mol } O_2$$

$$2 \text{ mol } H_2 = 1 \text{ mol } O_2$$

Mole Ratios Practice

- Given the following equation: $N_2 + 3H_2 \rightarrow 2NH_3$

Write each of the mole ratios.

Mole Ratios Practice 2

- Given the following equation: $N_2 + O_2 \rightarrow 2NO$

Write each of the mole ratios.

LA problem - 1

- Ethane (C_2H_6) burns when ignited in the presence of oxygen.

Mole–Mole Conversions Practice

- Given the following equation: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

How many moles of Fe can be produced from 17.5 mol of CO?

Mole–Mole Conversions Practice 2

- Given the following equation: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$

How many moles of O_2 are needed to react with 0.987 mol of C_2H_6?

Mole–Mole Conversions Practice 3

- Ammonia reacts with sodium hypochlorite to produce sodium hydroxide and chloramine gas (which is a very toxic gas).

- 1.32 moles of ammonia will produce how much of the gas?

Converting Moles to Mass

- We can use the balanced equation for a reaction to calculate the numbers of moles of reactants and products for a particular case.
  - However, moles represent numbers of molecules, and we cannot count molecules directly.
  - In chemistry we count by weighing.
  - We can use the molar mass to convert between moles and mass.

Steps for Calculating the Masses of Reactants and Products in Chemical Reactions

1. Balance the equation for the reaction.
2. Convert the masses of reactants or products to moles.
3. Use the balanced equation to set up the appropriate mole ratio(s).
4. Use the mole ratio(s) to calculate the number of moles of the desired reactant or product.
5. Convert from moles back to masses.

Stoichiometry

- The process of using a chemical equation to calculate the relative masses of reactants and products involved in a reaction is called stoichiometry.
- Chemists say that the balanced equation for a chemical reaction describes the stoichiometry of the reaction.

Think, Pair, Share Activity – Unit 9.3

- Turn to your neighbor and explain:
  - Why is it important to balance an equation before using it in a stoichiometry problem?

Stoichiometry Practice

- Given the following equation: $2SO_2 + O_2 \rightarrow 2SO_3$

How many grams of SO_3 can be produced from 2.33 mol of O_2?

Stoichiometry Practice 2

- Given the following equation: $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$

How many moles of H_2SO_4 are needed to react with 19.9 g of KOH?

Stoichiometry Practice 3

- Given the following equation: $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

How many grams of NH_3 are needed to produce 54.7 g of NO?

Stoichiometry Practice 4

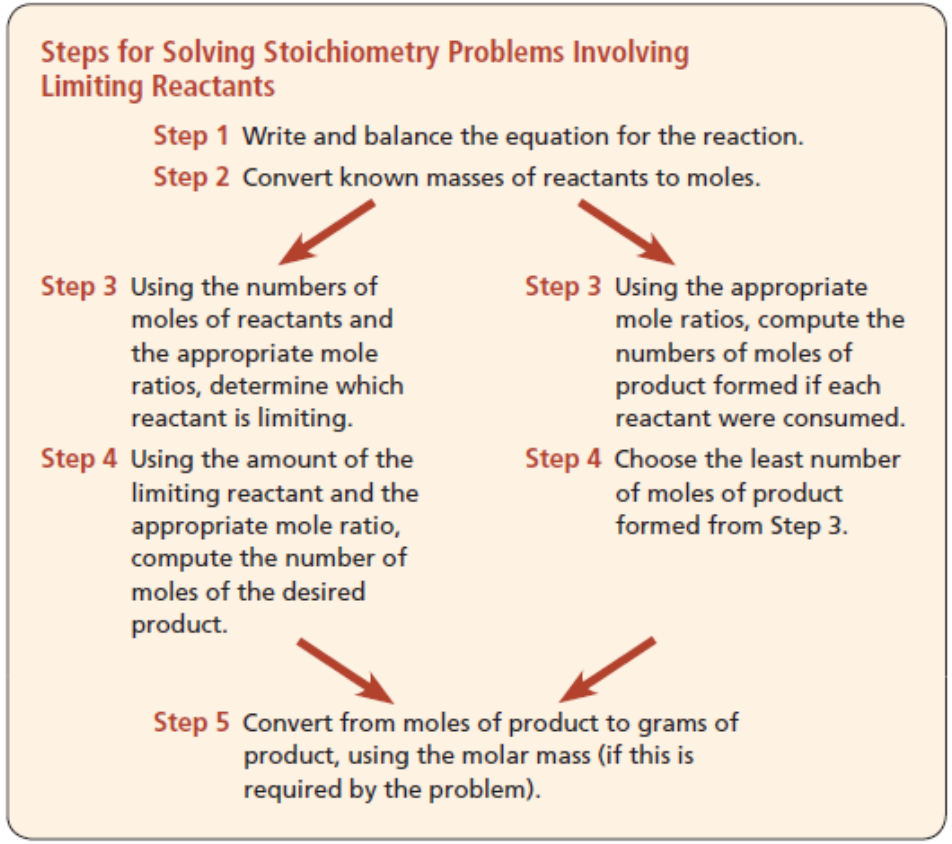
- Given the following equation: $CuSO_4 + Fe \rightarrow FeSO_4 + Cu$

How many grams of $CuSO_4$ are needed to react with 244 g of Fe?

Limiting Reactants

- In a chemical reaction, one reactant is generally used up first.
  - This reactant is known as the limiting reactant or limiting reagent.
  - The limiting reactant controls how much product is formed.
- To determine how much product can be formed from a given mixture of reactants, we have to look for the reactant that is limiting—the one that runs out first and thus limits the amount of product that can form.

Steps for Solving Stoichiometry Problems Involving Limiting Reactants



Limiting Reactants Practice

- Given the following equation: $3Mg + N_2 \rightarrow Mg_3N_2$

If 4.75 mol Mg reacts with 2.11 mol N_2, what is the limiting reactant?

Limiting Reactants Practice 2

- Given the following equation: $4Al + 3O_2 \rightarrow 2Al_2O_3$

If 17.9 g Al reacts with 64 g O_2, what is the limiting reactant?

Theoretical Yield

- In doing our previous calculations, we used the fact that the amount of product is controlled by the limiting reactant.
- Products stop forming when one reactant runs out.
- The amount of product calculated in this way is called the theoretical yield of that product.
  - It is the amount of product predicted from the amounts of reactants used.

Theoretical Yield Practice

- Given the following equation: $Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2 + 2NaNO_3$

If 237 g NaCl reacts with excess $Pb(NO_3)_2$, what is the theoretical yield of $PbCl_2$?

Theoretical Yield Practice 2

- Given the following equation: $2Na + Cl_2 \rightarrow 2NaCl$

If 68 g Na reacts with 97 g Cl_2, what is the theoretical yield of NaCl?

Actual Yield

- The theoretical yield is the maximum amount of product that can be produced from the quantities of reactants used.
- Actually, however, the amount of product predicted (the theoretical yield) is seldom obtained.
  - One reason for this is the presence of side reactions (other reactions that consume one or more of the reactants or products).
- The amount of product actually produced is known as the actual yield.

Think, Pair, Share Activity – Unit 9.6

- Turn to your neighbor and explain:
  - Is it possible to obtain an actual yield that is higher than the theoretical yield? Why or why not?

Percent Yield

- The actual yield of product, which is the amount of product actually obtained, is often compared to the theoretical yield.
- This comparison, usually expressed as a percentage, is called the percent yield.
- Percent yield is important as an indicator of the efficiency of a particular reaction.

$$\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\% = \text{percent yield}$$

Percent Yield Practice

- Given the following equation: $H_2 + I_2 \rightarrow 2HI$

If 69.1 g H_2 reacts with excess I_2 and 134 g HI are actually produced, what is the percent yield?

Percent Yield Practice 2

- Given the following equation: $2KClO_3 \rightarrow 2KCl + 3O_2$

If 191 g $KClO_3$ decomposes and 89.9 g KCl are actually produced, what is the percent yield?